HKUTG Quant Team Recruitment – Research Track

Background

You are given **data.csv**, fourteen years of daily data for a synthetic market — a market index, ten stocks, an ETF, and a listed option contract — ending on 2019-12-31, and **assignment.ipynb**, the notebook you will work in and submit. Your goal is to **predict three hidden targets** over a period you do not have, every Friday for the following week.

- **T1** — next week's return of e1 (70%)
- **T2** — next week's realized volatility of e2 (15%)
- **T3** — next week's return of the ETF (15%)

The notebook

The notebook runs end to end as given. It loads the data, builds daily and weekly representations of every instrument, defines the targets, provides a toolbox of analysis functions, fits a fixed least-squares model on whatever features you supply, and ends with a check that your code executes. Only the three **TODO** sections are yours.

Your tasks

- **TODO A — Exploration.** Work out what structure is in the data. Not scored, but read.
- **TODO B — Features.** One function per target, each returning **at most five predictors**. You do not choose the model, only what you give it.
- **TODO C — Validation.** Everything you were given is training data. Any out-of-sample evidence has to be carved out of the fourteen years yourself. Show it in code.

What we hope to see

- Evidence that you looked at the data before you modelled it.
- Features kept for a reason you can state, and features dropped for reasons other than a low score.
- Restraint — five slots per target, spent deliberately.
- Code another person can read and follow.

Rules

1. **No look-ahead.** A feature may only use data dated on or before its own prediction date. Violations are detectable and are treated as disqualifying.
2. **At most five features per target**, enforced by the notebook.
3. **Not all provided functions are useful.** The toolbox is deliberately larger than this problem needs. Using one is a decision, not a default.
4. **AI tools are allowed.** However, make sure that you understand the code and can justify the idea behind it. You may be asked about the code you submit in later rounds of the interview process.

How you are scored

We run your feature functions on the full data, which continues past 2019-12-31. They are fitted on the years you were given and applied, unchanged, to the years you were not. The quantitative metric is the **information coefficient** — the correlation between your predictions and what actually happened — combined by the weights above. Your notebook and written reasoning are assessed separately. The notebook explains the scoring in full.

Submission

1. **The notebook**, with the TODO sections filled in. There is no submission file — your predictions are produced by us, from your code.
2. **A separate write-up**, 400 words maximum across the four questions below.

1. What is in the data? Describe the structure you believe you found, and how you found it.
2. Why do your features carry information about their target?
3. What ideas did you reject, and why? Reasons other than a low score.
4. What would you do next, given more time?

Optional, not counted towards the word limit: did anything in the data look wrong, artificial, or otherwise strange to you? What did you do about it?

The notebook holds code only — every written answer belongs in the write-up. Good luck!